HIMANSHU SRIVASTAV

Software Development Engineer (SDE) | Python Backend (FastAPI, Django, Flask) & AI/ML Systems Engineer

Current Location: Udaipur, Rajasthan, India | **Preferred Locations:** Noida, Gurugram, Delhi NCR (Priority Relocation)

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EXECUTIVE SUMMARY

Software Development Engineer (SDE) and Python Backend & AI Engineer with 2+ years of production experience designing, testing, and debugging high-concurrency asynchronous backends, distributed architectures, and deep learning microservices. Proven track record of independently shipping 8+ end-to-end enterprise production systems (maintaining 500+ Python files across 8 repositories) spanning multi-tenant SaaS, biometric AI, real-time GPS tracking, legal-tech, and geospatial platforms. Expert in FastAPI, Django, Django REST Framework (DRF), Flask, SQLAlchemy 2.x (async), PostgreSQL 18, Celery, and Redis—highlighted by a solo zero-downtime database migration of 5.08M+ rows (slashing API latency from 5–10s to sub-second) and an 18-day full async framework rewrite. Deeply experienced in architecting standardized Deep Learning Bundles (DLBs), fine-tuning Vision Transformers and CNNs (RF-DETR, Detectron2, ResUNet, PyTorch) on NVIDIA Ada Lovelace GPUs (bfloat16 AMP), and orchestrating fault-tolerant workloads across AWS (EKS, S3, EC2, RDS, EFS), OCI, Azure, and GCP. Ready for immediate relocation to Noida, Gurugram, or Delhi NCR.

CORE TECHNICAL SKILLS

- **Languages & CS Fundamentals:** Python 3.10+, SQL, Java (OOP), JavaScript, HTML5, CSS3, Shell Scripting; Data Structures, Algorithms, Object-Oriented Programming (OOP) Design, Complexity Analysis, Query Optimization.
- **Backend & Web Frameworks:** FastAPI, Django, Django REST Framework (DRF), adrf (async DRF), Flask, SQLAlchemy 2.x (sync & async), Django ORM, Alembic, Django Channels, RESTful API Design, Microservices, WebSockets, Daphne / ASGI, Gunicorn, Uvicorn.
- **Databases, Caching & Message Queues:** PostgreSQL 18, MySQL, SQLite, aiomysql, Qdrant Vector DB, OpenSearch, Snowflake / Fabric DB syncing, Redis, Celery, background task scheduling; multi-tenant DB routing, schema redesign, N+1 query elimination, pgloader zero-downtime migrations (5.08M rows).
- **APIs, Authentication & Security:** REST, Webhooks, Meta WhatsApp Cloud API, MSG91 API v5, Traccar GPS API, Supabase, Python requests, Postman; OTP/JWT (HS256, simplejwt), OAuth, Two-Factor Authentication (2FA), Role-Based Access Control (RBAC), bcrypt, HMAC-SHA256 signed URLs.
- **Cloud, DevOps & Testing:** AWS (EKS, S3, EFS, EC2, RDS), OCI (Oracle Cloud), Azure (Blob, Functions), GCP, Linux (WSL2/Ubuntu/VPS); Docker, Docker Compose, Kubernetes, Nginx, Let's Encrypt SSL, Git, GitHub Actions CI/CD, GHCR, Python virtual environments (venv), automated test suites (pytest), unit/integration debugging, production incident triage under live load.
- **Deep Learning & Computer Vision:** PyTorch, TensorFlow, Keras, OpenCV, RF-DETR (Lightweight DETR + DINOv2 ViT Backbone), Detectron2 (Mask R-CNN, Faster R-CNN), ResUNet-4B, FCN8, SAM, Grounding-DINO, DeepFace (FaceNet), Silent-Face Anti-Spoofing (MiniFASNetV2), Hungarian Bipartite Loss, Soft Dice Loss, Transfer Learning, Anchor Optimization, mixed-precision computing (bfloat16/FP16 AMP) on NVIDIA RTX 5000 Ada, NumPy, Pandas, scikit-learn.
- **Geospatial, Scraping & Hardware:** GDAL, Rasterio, Shapely, GeoPandas, ArcGIS Python API (arcpy), geopy, STAC, GeoTIFF, Orthorectification, Cross-Tile NMS, Sliding-Window Tiling, Hanning-blend, NDVI; Playwright, Selenium, BeautifulSoup4, APScheduler; PyQt6 desktop GUIs, pyserial (physical RFID hardware integration).

WORK EXPERIENCE

GJ Maps Solutions Pvt. Ltd. | Udaipur, Rajasthan **June 2026 – Present**

Software Developer / Python Deep Learning & Backend Engineer

- **Xenterra-Fabric (Distributed Backend Platform):** Architected, wrote, tested, and debugged production Python REST APIs for OAuth and 2FA authentication, billing/subscription engines (invoicing, credit ledgers, AWS-based pricing), workspace management, and CRUD endpoints backed by PostgreSQL sessions, Redis, and background task queues.
- **Xenterra-Pulse & Deep Learning Bundles (DLBs):** Designed and deployed standardized DLBs containing deterministic runtime specifications (`metadata.json`), catalog schemas (`data.json`), and isolated execution engines (`inference/main.py`) validated via automated `pytest` suites.
- **Fault-Tolerant Cloud Orchestration:** Eliminated silent runtime failures and platform hanging bugs by replacing mock fallbacks with strict fail-loud exception contracts and non-blocking job progress reporting (`update_job_status`) across distributed Kubernetes/EKS worker pods.
- **MLOps Packaging & Multi-Cloud Sync:** Built a dual-mode execution interface supporting cloud task payloads (AWS EFS/POSIX path normalization) and CLI debugging, alongside custom packaging utilities (`create_dlb.py`) maintaining lean 83–156 MB bundles; synchronized Snowflake/Fabric and Pulse DB catalogs with automated S3 artifact deployment, ArcGIS/STAC integration, and OCI/Azure CI/CD pipelines.
- **Geospatial AI & Satellite Delineation:** Built a continental agricultural field delineation pipeline with Detectron2 Mask R-CNN (ResNet-50-FPN) on 1.0m USDA NAIP imagery, achieving 68.84% Mask AP and 70.24% Box AP across

10,178 image chips and 4.8M vector annotations.

- **Smallholder Parcel Engine & GPU Optimization:** Co-developed a 5.0m cadastral parcel engine for India (ISRO LISS-4 4-band rasters, 146k instances) with custom RPN anchors (`[[12], [24], [48], [96], [192]]`); trained for 50,609 iterations (30 epochs) using bfloat16 AMP on an NVIDIA RTX 5000 Ada (32GB VRAM) at a steady ∼3.8GB VRAM footprint and ∼3.2 batches/sec.
- **High-Throughput Sliding-Window Inference & ResUNet-4B:** Engineered a 512 × 512 sliding-window pipeline (64px overlap) with cross-tile spatial NMS deduplication, delineating 2,589 farm parcels in 15.5s (∼20 FPS) into WGS84 GeoJSON polygons; built a 4-band ResUNet-4B boundary segmenter with Compound Boundary Loss, dynamic NDVI normalization, and 2D Hanning-window blending.
- **Vision Transformers (RF-DETR Medium, 33.4M Params, DINOv2 Backbone):** Upgraded legacy Faster R-CNN and YOLO pipelines across three domains: (a) *License Plate Anonymization:* aggregated 37,830 images / 45,716 annotations (Kaggle, Hugging Face, 2,078 Goa street scenes) with S3 ingestion and Grounding-DINO labels, reaching 97.64% mAP@50 and 72.13% strict mAP@50:95; (b) *Windmill Blade Defects:* achieved 87.30% mAP@50 and 85.77% F1 across 2,995 DTU Nordtank drone frames; (c) *Maritime SAR Ship Detection:* reached 59.24% mAP@50, 75.00% precision, and sub-2px localization on LS-SSDD SAR GeoTIFFs, halting at Epoch 13 to prevent overfitting.
- **Foundation Model Standardization:** Resolved DETR zero-indexed foreground mappings, CPU worker fallbacks, and GSD mismatches; packaged Microsoft Global Renewables Watch (FCN8 ResNet-50) and NASA-IBM Prithvi foundation models (Burn Area, Flood, Crop ID) into production DLBs.

MR Softwares | Ujjain, Madhya Pradesh **May 2025 – June 2026**
Python Full Stack Developer

- Independently designed, coded, tested, and debugged 8 end-to-end production applications across 500+ Python files—owning database schemas, REST APIs, Git workflows, virtual environments, Docker Compose, Nginx + Let's Encrypt SSL, and multi-cloud CI/CD across OCI, Azure, and Linux VPS.
- **WaahApp (WhatsApp Business SaaS — Django, Celery, Redis, Django Channels, PostgreSQL 18):** Built a multi-tenant real-time WebSocket and Meta WhatsApp Cloud API webhook platform; independently executed a zero-downtime MySQL to PostgreSQL 18 and Azure to OCI migration of 5.08M rows in 2h 15m with zero data loss, tuning `shared_buffers` and indexes to slash API latency from 5–10s to sub-second.
- **eSchool Biometric & AI Attendance (Django+DRF to FastAPI Async):** Led a complete async framework rewrite using `aiomysql` and SQLAlchemy 2.x async connection pooling in 18 working days with zero downtime; reprogrammed multi-tenant DB routing and integrated DeepFace (FaceNet) embeddings, sub-second Qdrant cosine matching, and Silent-Face Anti-Spoofing (MiniFASNetV2 PyTorch).
- **Avvyx (B2B SaaS PPF Warranty Platform — FastAPI, SQLAlchemy 2.x, Alembic, PostgreSQL):** Architected full backend from scratch with 11 ORM models, OTP+JWT (HS256) auth, 5-step registration workflow, programmatic A4 PDF certificate and QR code generation, RBAC, audit logging, and GitHub Actions CI/CD auto-deploying to GHCR behind Nginx + SSL.
- **LatLong GPS Microservice & API Integrations (Django/DRF, Traccar, OpenSearch):** Built distributed middleware connecting Traccar GPS and OpenSearch with `geopy` geodesic-radius proximity alerts and HMAC-SHA256 signed tracking URLs, plus production webhook integrations for MSG91 SMS API v5 and Supabase.
- **Production Debugging & ORM Optimization:** Diagnosed and fixed a critical live billing bug (`EnumChoiceField` integer mismatch silently zeroing client wallets) under live load with zero data loss; eliminated N+1 ORM bottlenecks via `select_related`, `prefetch_related`, and raw SQL aggregation.
- **SCS Legal Scraper & Hardware Systems:** Built a multi-court scraping engine across 6+ High Courts (Playwright, Selenium, BeautifulSoup4, APScheduler) deployed on Azure Functions + Docker (cutting manual data collection time by ∼80%), and a PyQt6 desktop hotel management system interfacing with physical RFID card writers via `pyserial`.

CodeAlpha | Remote **October 2023 – January 2024**
Python Developer Intern

- Wrote, tested, and debugged reusable Python automation scripts for data-processing pipelines, reducing manual workflow time by ∼15% and delivering production-ready data transformation utilities for live reporting systems.

Oasis Infobyte | Remote **April 2023 – May 2023**
Software Engineer Intern

- Shipped 3 client-facing web applications within a 30-day deadline using Python, HTML5, CSS3, and JavaScript under ambiguous requirements.

KEY PRODUCTION METRICS & ARCHITECTURAL HIGHLIGHTS

System / Domain	Stack & Architecture	Verified Production Impact & Metrics
WaahApp SaaS	Django, Channels, Celery, Redis, PostgreSQL 18, OCI	**5.08M rows** migrated in **2h 15m** with **0 data loss**; API latency cut from **5–10s to <1s**.
eSchool Biometrics	FastAPI, SQLAlchemy 2.x Async, aiomysql, Qdrant, PyTorch	Full async rewrite from Django+DRF in **18 working days** with **0 downtime**; sub-second vector face matching.
Avvyx B2B SaaS	FastAPI, SQLAlchemy 2.x, Alembic, PostgreSQL, Docker, GHCR	Built **11 ORM models**, 5-step onboarding, OTP/JWT auth, RBAC, and programmatic A4 PDF/QR generation from scratch.
Xenterra Platform	FastAPI/Django, PostgreSQL, Redis, EKS, S3, Snowflake, DLBs	Standardized **83–156 MB** `.dlb` bundles; OAuth/2FA, credit-ledger billing APIs, and fail-loud EKS worker telemetry.
Geospatial AI	Detectron2, ResUNet-4B, GDAL, Rasterio, RTX 5000 Ada	**68.84% Mask AP** (4.8M vectors); **2,589 parcels in 15.5s (∼20 FPS)**; steady **∼3.8GB VRAM** over 50,609 iterations.
RF-DETR Suite	RF-DETR Medium (DINOv2), Grounding-DINO, SAR GeoTIFFs	**97.64% mAP@50** (37.8k license plate images); **87.30% mAP@50** (windmill defects); **75.00% precision** (SAR ships).

EDUCATION & CERTIFICATIONS

- **B.Tech in Computer Science & Information Technology** — IES IPS Academy, Indore, MP | *2021 – 2025* | **GPA: 7.50 / 10.0**
- **Senior Secondary (Class XII), Science (PCM)** — RPM Academy, Gorakhpur, UP | *2019 – 2021*
- **Certifications:** Python Developer Associate (PDAC-24) — ITechCert / AICERT (*Jul 2024*); Computer Vision Builder — IIT Delhi, FITT / MP-IITD (*Mar 2024*).